Math 110 — Week 6

Solutions

HW6 P1.

Idea. An equation satisfied by P is satisfied by its eigenvalues. For (b), Pv is the range part of v and what is left is in the null space.

1. (a) Let $Pv = \lambda v$, $v \neq 0$. Then $P^2v = P(\lambda v) = \lambda^2 v$. Since $P^2 = P$, $\lambda^2 v = \lambda v$, so $(\lambda^2 - \lambda)v = 0$, so $\lambda(\lambda - 1) = 0$.
2. (b) *Sum.* For any v: $v = (v - Pv) + Pv$. Here $Pv \in \operatorname{range} P$, and $P(v - Pv) = Pv - P^2v = 0$, so $v - Pv \in \operatorname{null} P$.
3. *Direct.* Let $v \in \operatorname{null} P \cap \operatorname{range} P$. Then $v = Pw$ for some w, and $Pv = 0$. So $v = Pw = P^2w = P(Pw) = Pv = 0$.

HW6 P2.

Idea. Differentiation lowers the degree of a nonconstant polynomial, and a nonzero multiple of p has the same degree as p.

1. (a) Suppose $p' = \lambda p$ with $p \neq 0$. If $\lambda \neq 0$, then $\deg(\lambda p) = \deg p$, but $\deg p' < \deg p$ (or $p' = 0$ when p is constant, while $\lambda p \neq 0$). Contradiction. So $\lambda = 0$.
2. For $\lambda = 0$: $p' = 0$ means p is constant. So the only eigenvalue is 0, and $E(0, D) = \operatorname{span}(1)$.
3. (b) On smooth functions, $e^{\lambda x}$ satisfies $(e^{\lambda x})' = \lambda e^{\lambda x}$, so every $\lambda \in \mathbb{R}$ is an eigenvalue. The degree argument is not available, and $e^{\lambda x}$ is not a polynomial for $\lambda \neq 0$. (This space is infinite-dimensional, so having infinitely many eigenvalues contradicts nothing.)

HW6 P3.

Idea. $S^{-1}TSv = \lambda v$ becomes $T(Sv) = \lambda(Sv)$ after applying S.

1. $v \in E(\lambda, R) \iff S^{-1}TSv = \lambda v \iff TSv = \lambda Sv$ (apply S; reversible because S is invertible) $\iff Sv \in E(\lambda, T) \iff v \in S^{-1}E(\lambda, T)$.
2. S^{-1} is an isomorphism, so $E(\lambda, R) \neq \{0\}$ exactly when $E(\lambda, T) \neq \{0\}$. So R and T have the same eigenvalues (and the eigenspaces have the same dimensions).

HW6 P4.

Idea. Expand; then eigenvalue $\iff \lambda I - A$ not invertible $\iff$ determinant zero.

1. $tI - A = \left(\begin{smallmatrix} t-a & -b \\ -c & t-d \end{smallmatrix}\right)$, so $\det(tI - A) = (t-a)(t-d) - (-b)(-c) = t^2 - (a+d)t + ad - bc$.
2. λ is an eigenvalue $\iff (A - \lambda I)x = 0$ has a nonzero solution $\iff \lambda I - A$ is not injective $\iff \lambda I - A$ is not invertible (operator on $\mathbb{F}^2$) $\iff \det(\lambda I - A) = 0$ (Week 4 criterion).

HW6 P5.

Idea. $t^2 - (a+d)t + (ad - bc)$, then one row of $A - \lambda I$.

1. A_1: $\chi = t^2 + 1$. Over $\mathbb{R}$: no roots, no eigenvalues, no eigenvectors. Over $\mathbb{C}$: $\lambda = \pm i$. $A_1(x, y) = (-y, x)$. For $\lambda = i$: $-y = ix$, so $E(i) = \operatorname{span}\big((1, -i)\big)$. For $\lambda = -i$ (conjugate): $E(-i) = \operatorname{span}\big((1, i)\big)$. Two distinct eigenvalues in dimension 2: a basis of eigenvectors exists over $\mathbb{C}$.
2. A_2: $\chi = t^2 - 2t + 1 = (t-1)^2$. Only $\lambda = 1$. $(A_2 - I)(x, y) = (y, 0) = 0$ gives $y = 0$: $E(1) = \operatorname{span}(e_1)$, dimension 1. Every eigenvector is a multiple of e_1, so there is no basis of eigenvectors.
3. A_3: $\chi = t^2 - t - 1$, $\lambda_\pm = \frac{1 \pm \sqrt{5}}{2}$. Eigenvector $(b, \lambda - a) = (1, \lambda)$: check $A_3(1, \lambda) = (\lambda, 1 + \lambda) = (\lambda, \lambda^2)$ since $\lambda^2 = \lambda + 1$. $E(\lambda_\pm) = \operatorname{span}\big((1, \lambda_\pm)\big)$. Two distinct real eigenvalues: a basis of eigenvectors exists.

HW6 P6.

Idea. Applying T shifts each $T^k v$ to $T^{k+1}v$. Once one power depends on the earlier ones, all later powers do.

1. (a) An element of $C(v,T)$ is a finite combination $\sum_k a_k T^k v$. Applying T gives $\sum_k a_k T^{k+1} v \in C(v,T)$.
2. (b) Let $n = \dim V$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so there is a smallest m with $T^m v \in \operatorname{span}(v, \ldots, T^{m-1}v) =: U$; $m \geq 1$ because $v \neq 0$, and $m \leq n$.
3. U is T-invariant: T sends $T^k v$ to $T^{k+1}v \in U$ for $k < m-1$, and $T^{m-1}v$ to $T^m v \in U$.
4. So $T^k v \in U$ for all k (induction: $T^{k+1}v = T(T^k v) \in T(U) \subseteq U$). Hence $C(v,T) \subseteq U$, and $U \subseteq C(v,T)$ is clear from the definition.

HW6 P7.

Idea. Solve $(P - I)p = 0$ and normalise. Here the rows also sum to 1, so the uniform vector works.

1. $Pp = p$: $\frac{1}{2}p_1 + \frac{1}{2}p_2 = p_1$ gives $p_2 = p_1$; $\frac{1}{2}p_2 + \frac{1}{2}p_3 = p_3$ gives $p_3 = p_2$.
2. With $p_1 + p_2 + p_3 = 1$: $p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Check the middle row: $\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{3}$.
3. Meaning: if the chain's state is uniformly distributed at one time, it is uniformly distributed at every later time; in the stationary regime each state is occupied one third of the time.

R1.

Idea. Powers first, then combine.

1. $T^k v = \lambda^k v$ for all $k \geq 0$: true for $k = 0$, and $T^{k+1}v = T(\lambda^k v) = \lambda^k T v = \lambda^{k+1} v$.
2. For $p(z) = \sum a_k z^k$: $p(T)v = \sum a_k T^k v = \sum a_k \lambda^k v = p(\lambda)v$.

R2.

Idea. Use $(T'\varphi)(u) = \varphi(Tu)$.

1. If $u \in \operatorname{null} T$ then $Tu = 0 \in \operatorname{null} T$. If $u \in \operatorname{range} T$ then $Tu \in \operatorname{range} T$, being an output of T.
2. Let $\varphi \in (\operatorname{null} T)^0$. For $u \in \operatorname{null} T$: $(T'\varphi)(u) = \varphi(Tu) = \varphi(0) = 0$. So $T'\varphi \in (\operatorname{null} T)^0$.
3. Let $\varphi \in (\operatorname{range} T)^0$. For any v: $(T'\varphi)(v) = \varphi(Tv) = 0$ because $Tv \in \operatorname{range} T$. So $T'\varphi = 0$, which lies in $(\operatorname{range} T)^0$.

R3.

Idea. Look for the smallest counterexample.

(a) False. For $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, e_1 and e_2 are eigenvectors, and $e_1 + e_2 \mapsto (1,2)$ is not a multiple of $(1,1)$. (Also $v_1 + v_2$ could be 0.) (b) False. Rotation by $90°$ on $\mathbb{R}^2$ has no eigenvector.

E1.

Idea. Three linear equations; substitute.

1. $P = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{smallmatrix}\right)$, $\alpha = \frac{1}{2}$, $(1-\alpha)v = (\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$.
2. Equations: $r_1 = \frac{1}{2}r_3 + \frac{1}{6}$, $\quad r_2 = \frac{1}{4}r_1 + \frac{1}{6}$, $\quad r_3 = \frac{1}{4}r_1 + \frac{1}{2}r_2 + \frac{1}{6}$.
3. Substitute r_2 into the third: $r_3 = \frac{1}{4}r_1 + \frac{1}{8}r_1 + \frac{1}{12} + \frac{1}{6} = \frac{3}{8}r_1 + \frac{1}{4}$. Then $r_1 = \frac{3}{16}r_1 + \frac{1}{8} + \frac{1}{6}$, so $\frac{13}{16}r_1 = \frac{7}{24}$ and $r_1 = \frac{14}{39}$.
4. $r_2 = \frac{14}{156} + \frac{26}{156} = \frac{10}{39}$, $r_3 = \frac{15}{39}$. So $r = \frac{1}{39}(14, 10, 15)$; the coordinates sum to 1. Ranking: page 3, then 1, then 2.